SE Lab 7 — Fuzz Testing

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Objective

Use Hypothesis-based fuzz testing to explore unexpected inputs in the museum admission parser, identify input-validation failures, and update the application to handle invalid input safely.

Tools

Python • Hypothesis • pytest

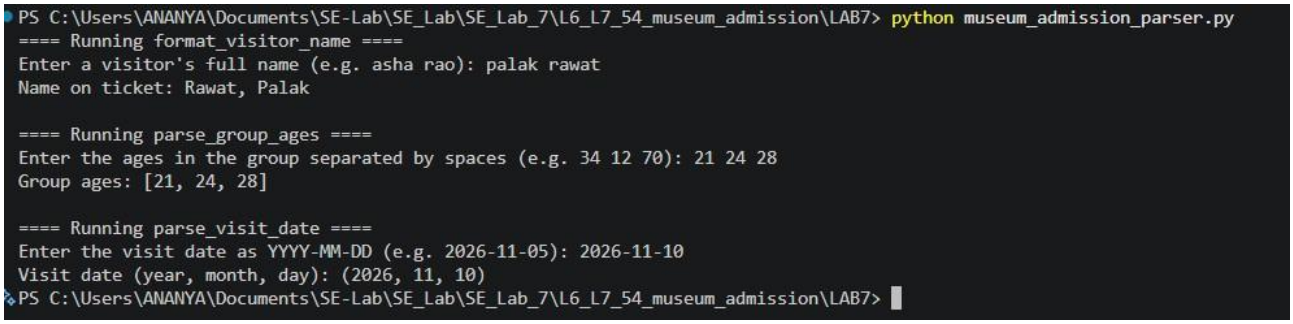
Functions tested

1. `format_visitor_name(name)` — formats a visitor's name as Surname, Given names.
2. `parse_group_ages(text)` — parses space-separated ages into integers.
3. `parse_visit_date(text)` — parses a YYYY-MM-DD date into a (year, month, day) tuple.

Testing approach

Hypothesis generated a variety of text and None inputs for each function. The initial tests exposed failures for invalid inputs. The parser was then updated with input validation and exception handling. The final test run completed successfully.

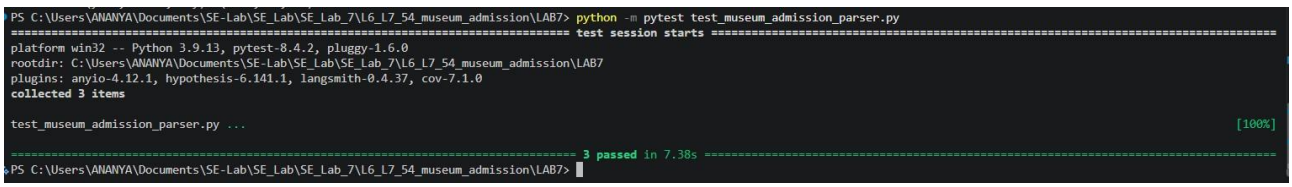
Original Application Output



Failing Hypothesis Tests



Passing Hypothesis Tests



Fixed Application Output

